AYMANE AIT DADS

AI & Deep Learning Intern | Radar Classification · Real-Time Detection · Deep Learning Research

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams (score 0.86/1.0, $106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a 30B-parameter model with LoRA SFT and staged continuation training - directly demonstrating research ownership from literature-informed design through implementation and evaluation. Ranked 1st in EURECOM class leaderboard for AI-generated image detection (0.791 AUC) using CLIP ViT-L/14 fine-tuning with ensemble methods and 10-view TTA, and submitted to NTIRE 2026 @ CVPR CodaBench international benchmark. Brings hands-on deep learning research in computer vision, signal-domain classification, and transformer fine-tuning - skills that map directly to intelligent radar target classification and real-time detection systems. Aims to contribute to next-generation radar tracking research at Robin Radar by applying ablation-driven experimentation and custom evaluation pipelines to drone and bird classification challenges.

CORE COMPETENCIES

Deep Learning Research | Real-Time Detection | Classification & Tracking | Ablation Studies | Computer Vision | Transformer Fine-Tuning | Literature Review | Evaluation Pipelines

WORK EXPERIENCE

Orange Maroc — *Jul 2024 - Aug 2024*

Data Science Intern - Casablanca, Morocco

- Built **Random Forest + K-Means classification pipeline** on 100,000+ fiber-optic network records (upload speed, latency, jitter) to classify network quality across thousands of nodes.
- Developed clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for operational decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and structured reporting - accelerating the team's diagnostic turnaround.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, bridging technical results and business decision-making for non-technical audiences.

PROJECTS

AI-Generated Image Detection - EURECOM ImSecu + NTIRE 2026 @ CVPR — *EURECOM Kaggle Leaderboard · 2025 · Ranked 1st in Class · CVPR Submission*

- Ranked **1st in EURECOM class** private Kaggle leaderboard (0.791 AUC) fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates, and progressive block unfreezing across 250K training images from 25+ generator types.
- Discovered that **1-epoch training outperformed 4-epoch (0.793 vs. 0.767 AUC)** for out-of-distribution generalization; built a 3-model ensemble with validation-AUC weighting and 10-view TTA, then submitted to NTIRE 2026 @ CVPR CodaBench international benchmark.

PyTorch · OpenCLIP · ViT · AdamW · FP16 Mixed Precision

NVIDIA Nemotron Model Reasoning Challenge - Kaggle (In Progress) — *Kaggle Competition · June 2026 · $106K+ Prize Pool · In Progress*

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter** Mamba-Transformer with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built a diagnostic evaluation pipeline revealing that reasoning over-expansion - not incorrect reasoning - was the primary failure mode, guiding targeted data curation with custom cipher parsers and 8-bit operation solvers.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Triton

Anomalous Sound Detection - Industrial Equipment — *EURECOM · 2025 · Unsupervised Fault Detection*

- Designed an unsupervised fault detection system for slide-rail machinery using a **Transformer autoencoder on Mel spectrograms with SpecAugment**, achieving AUC 0.80 with no labeled anomaly data.
- Framed the task as a **signal-domain classification problem** under real-world constraints - no anomaly labels, variable noise - directly mirroring the challenges of radar target classification in cluttered environments.

PyTorch · Transformer Autoencoder · Mel Spectrograms · SpecAugment

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM — *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Image Security, Statistics, Cloud Computing

CPGE - Mathematics & Physics - Ibn Timiya — *2021 - 2023*

SKILLS

Deep Learning & Computer Vision: PyTorch | CLIP ViT | CNN | DenseNet | Transformer Fine-Tuning | FP16/BF16 Mixed Precision | Ensemble Methods

LLM Fine-Tuning & Research: LoRA | PEFT | SFTTrainer | Unsloth | vLLM | Hugging Face Transformers | Ablation Studies

ML Engineering & Data Science: Python | Scikit-learn | Pandas | NumPy | Feature Engineering | Git | Kaggle GPU | Power BI